



NITHISH RAVIKKUMAR

Roorkee, Uttarakhand, India - 247667

nithish232005@gmail.com

[Github](#)

+91-9205256782

EDUCATION

- **Indian Institute of Technology Roorkee (IITR)**, Roorkee, Uttarakhand
B.Tech in Engineering Physics

Aug 2023 – 2027

TECHNICAL SKILLS

- **Programming Frameworks:** Python, R, SQL, PyTorch, TensorFlow, Scikit-Learn, LangChain, LangGraph, Hugging Face, NumPy, Pandas, SciPy, Matplotlib, Seaborn
- **Robotics & Hardware:** ROS2, Gazebo, Mavlink, Ardupilot, C++ (Arduino, ESP32), Altium Designer
- **Developer Tools:** Git, Docker, Google Cloud Platform (GCP), OpenCV, Pillow, Whisper, FreeSwitch, ONNX

EXPERIENCE

AI Developer Intern

Aviostack Pvt. Ltd.

Remote

May 2025 – June 2025

- Prototyped and deployed a **audio-modal AI agent** as Call Center Assistant for micro-warehouse E-commerce Deliveries.
- Developed end-to-end pipeline integrating **STT / TTS, VAD**, intent based classification and **RAG based retrieval** with **Telephony** services, enabling real-time query resolution with $\leq 0.8s$ response time latency.
- Optimized **cloud deployment (GCP)** using open-source LLM and speech components, along with **rule based-routing** to reduce token usage, cutting operating costs by over **60%** compared to commercial API solutions.

PROJECTS

- **Optical Flow Estimation and Collision Prevention on AMD Kria FPGA** March 2025 – Sept 2025
[Optical Flow estimation](#)
 - Designed a collision prevention system for autonomous vehicles using **PWC-Net** based optical flow and **YOLOv9T**.
 - Developed **Time-To-Collision (TTC) estimator** using divergence and flow magnitude-based heuristics, and **Echo State Network (ESN)**, enabling accurate trajectory prediction up to **300ms** ahead at **30 FPS**.
 - Applied quantization, pruning; compiled with Vitis AI for deployment on DPU, with **72%** reduction in inference time.
 - *Under Prof. Sparsh Mittal, ECE department, IITR for AMD OpenHardware Contest 2025*
- **Bias Detection and Ground Truth Inference in Media Text** Sept 2025 - Oct 2025
[Ground Truth Inference](#)
 - Designed a bias-neutralization pipeline using **BERT** encoder for token-level bias detection and **BART** based **seq2seq** editor, trained on Wiki Neutrality Corpus + multi-source news.
 - Achieved bias reduction from **0.78** $\rightarrow$ **0.31** while preserving meaning (**SBERT semantic similarity** = **0.84**).
 - Evaluated model on entailment consistency, achieving **85% factual entailment retention** and consistency using NLI-based entailment checks across left/center/right sources.
- **Warehouse Automation** Sept 2024 – Oct 2024
[FLIPKART GRID - Textual Extraction from Image](#)
 - Developed pipeline for automatic package sorting in a warehouse with **NLP** techniques and **Image Segmentation**.
 - Performed image preprocessing, used **OCR** and entity recognition with segmentation and text parsing using fine-tuned Small Language Models (**SLMs**) - NuExtract, Gemma-2B for text extraction from printed receipts.
 - Achieved **90%+** entity extraction accuracy for customer data from noisy, dynamic scanned images.
- **Document Distillation on Edge Compute** July 2025
[Document Distillation](#)
 - Architected document distillation system to ingest, structure, and extract ranked insights from large PDF collections based on user-defined needs on optimized **CPU** runtime.
 - Implemented a hybrid retrieval pipeline (Lexical search, **BM25** & **Sentence Transformer**) and applied **INT8 static quantization** via **ONNX Runtime**, achieving a **75%** model size reduction and a **3x** inference speedup.
 - Achieved execution time of $\leq 10s$ per document and multi-document summarization under $\leq 60s$.

- **Financial Time Series Forecasting** April 2025
 [Financial Time Series Forecasting](#)
 - Developed a DL pipeline to forecast market capitalization growth rates of Indian companies using historical indicators.
 - Trained and benchmarked predictive models including **ElasticNet**, **Decision Trees**, and **LSTM**; with best performing model (**LSTM** variant) achieved **96.4%** prediction accuracy.
 - Performed sectoral risk segmentation using volatility clustering and trend analysis to derive investment insights.
- **Optimization of Terrain Mapping and Drone navigation algorithms** Jan 2025
 [iRRT Path Planning](#)
 - Improved on **Informed-RRT** (IRRT) path planning framework in Python utilizing **Octree-based spatial partitioning** to achieve collision-free navigation across 3D LiDAR point cloud recreation of actual terrain .
 - Implemented **Elastic band Smoothing** and terrain-aware cost function for navigation, penalising high-gradient zones and collision-prone paths, improving efficiency and safety of path.
 - Developed for ISDC'25; secured **International Rank 5**.
- **Monte Carlo Simulation of Nanofilm deposition** Oct 2025
 [Nanofilm Deposition](#)
 - Built a Monte Carlo simulation framework in Python modelling nanofilm growth across various **stochastic models** — Random Deposition, Edwards-Wilkinson, KPZ, Ballistic Deposition and BKL algorithm.
 - Simulated temperature-dependent surface diffusion with Arrhenius kinetics and Metropolis acceptance in 1D and 2D.
 - Validated simulated growth exponents against theoretical scaling predictions across all models and temperature regimes.
- **Atmospheric Temperature Profiling** March 2026 - April 2026
 [Atmospheric Temperature Profiling](#)
 - Built a multi-source atmospheric analysis pipeline fusing radiosonde balloon soundings with **ERA5 reanalysis NetCDF** data, processing geospatial datasets using **xarray** and **SciPy** across four seasonal profiles.
 - Generated Skew-T Log-P diagrams with parcel overlays and computed thermodynamic indices (CAPE, CIN, LCL, LFC) by stitching observed and satellite-derived data.
- **Hand Gesture Controlled Rubik's Cube** Oct 2023
 [Air Cube](#)
 - Developed virtual Rubik's Cube game with **MediaPipe** for hand tracking and **OpenCV** for video processing to translate gestures into cube rotations.
 - Implemented **Kociemba** algorithm for an auto-solve feature and created a dynamic UI with **VPython** and Tkinter.
 - *Secured 2nd position in Syntax Error-2023, a nation-wide hackathon organised by SDS Labs, IITR.*

EXTRACURRICULARS

Robocon (Core Member) – Electronics and Robotics March 2024 – March 2025

- Optimized terrain mapping and **path planning** using **ML-based altitude-aware cost function** for Rover using Drone imaging - Secured **International Rank 5** in ISDC'25 (International Space Drone Challenge).
- Built and deployed DL models for **Real-time object detection** and engineered **Communication protocols** with integrated control systems for **Mars Rover** prototype, to enable **autonomous navigation** in robotic systems. - Secured **International Rank 10** in IRC'25 (International Rover Challenge)

ARIES (Core Member) – Robotics & Drones March 2024 – Feb 2025

- Simulated and tested UAV prototypes to tune flight control algorithms.
- Participated in and organized drone contests, such as **MathWorks Minidrone Challenge** hosted by IISc Bangalore.

NSS IIT Roorkee (Volunteer) – Editorial Cell October 2023 – April 2025

- Contributed to enhancing engagement and outreach by curating content for awareness campaigns and social initiatives.
- Managed the NSS social media presence, increasing visibility through content creation and public relations efforts.

ACHIEVEMENTS

- Secured **International Rank 10** in International Rover Challenge (IRC) 2025.
- Secured **International Rank 5** in International Space Drone Challenge (ISDC) 2025.
- **Semi-finalists** in **FLIPKART GRID 6.0** - Robotics Challenge, for designing Warehouse Automation system.
- **Runner-up** in Syntax Error X 2023, a **National level hackathon**.
- Secured **AIR 1992** in JEE Mains 2023 (among top **0.2%** of candidates) , **AIR 3282** in JEE Advanced 2023.
- Secured **AIR 178** in KVPY SA and NTSE Scholar.